# Glossary

***definitions of exotic terms***

**absolute.** Independent of one's reference frame; the same as measured in each and every reference frame.

**absolute horizon.** The surface of a black hole. See *horizon.*

**absolute space.** Newton's conception of the three-dimensional space in which we live as having a notion of absolute rest, and as having the property that the lengths of objects are independent of the motion of the reference frame in which they are measured.

**absolute time.** Newton's conception of time as being universal, with a unique, universally agreed upon notion of simultaneity of events and a unique, universally agreed upon time interval between any two events.

**accelerated observer.** An observer who does not fall freely.

**accretion disk.** A disk of gas that surrounds a black hole or neutron star. Friction in the disk makes the gas gradually spiral inward and *accrete* onto the hole or star.

**adiabatic index.** Same as *resistance to compression.*

**aether.** The hypothetical medium which (according to, nineteenth-century thinking) oscillates when electromagnetic waves go by, and by its oscillations, makes the waves possible. The aether was believed to be at rest in absolute space.

**angular momentum.** A measure of the amount of rotation that a body has. In this book the word *spin* is often used in place of "angular momentum."

**antimatter.** A form of material that is "anathema" to ordinary matter. To each type of particle of ordinary matter (for example, an electron or proton or neutron) there corresponds an almost identical antiparticle